

## Chapter 7 — Class 2

Property rights, bargaining, and types of goods

### Instructor preparation

**Length:** 55 minutes. **Slides:** 16–28. **Reading:** Chapter 7 notes, “Property Rights and Incentives” through “Exercise — Rights and Types of Goods.”

**Learning goals:** students can explain how ownership changes incentives, identify an efficient bargain under alternative rights, distinguish efficiency from distribution, and classify goods using rivalry and excludability.

**Board setup:** left side for a cost comparison and bargaining payments; right side for the two-by-two goods matrix. Keep “Who can be excluded?” and “Does one person’s use reduce availability for others?” visible.

### 0–5 minutes | Retrieve last class

Ask students to answer without notes:

1. “Why can private equilibrium differ from the efficient quantity?”
2. “For a negative externality, which cost is missing?”
3. “Must government always be the party that changes incentives?”

**Answers:** third-party effects may be omitted; marginal external cost; not always—affected parties may negotiate when rights are clear and bargaining is feasible.

**Say:** “Today we ask who controls a resource and who can benefit from it. These rules affect decisions.”

### 5–11 minutes | Ownership and incentives — slides 16–17

Use a privately owned bicycle. Write four verbs: **maintain, protect, conserve, trade.**

Ask for one action under each verb. Expected examples: repair brakes, lock the bicycle, avoid unnecessary wear, sell it to someone who values it more.

**Say:** “An owner receives many benefits of care and bears many costs of neglect. That creates an incentive to care for the resource.”

Then ask: “If I own my car, do I automatically consider all the delay I impose on other drivers?”

**Answer:** no. Ownership can improve care while leaving external effects unresolved.

Define **property rights** as rules specifying who can control or use a resource. Rights need to be clear and enforceable to guide a bargain.

## 11–27 minutes | Board example 1: cattle and crops — slides 18–20

Use about 3 minutes for the resource comparison, 8 for the two rights cases, and 5 for the expensive-fence variation.

**Write:**

Annual crop damage without fence = \$1,000

Annual fence cost = \$600

Assume the fence prevents all damage, cattle production continues, no cheaper remedy exists, and bargaining has no cost. All figures are annual so they can be compared directly.

### Step 1: find the efficient physical outcome

| Outcome  | Fence cost | Crop damage | Total resource cost |
|----------|------------|-------------|---------------------|
| No fence | 0          | 1,000       | 1,000               |
| Fence    | 600        | 0           | 600                 |

**Ask:** “Which outcome uses fewer resources?” **Answer:** build the fence; net gain is  $1,000 - 600 = 400$ .

**Say:** “First choose the lower-cost outcome. Then ask who pays.”

### Step 2: farmer has the right to undamaged crops

The rancher must prevent or compensate the damage. Compare paying 1,000 in compensation with building the 600 fence. The rancher chooses the fence.

**Say:** “The rancher bears the cost of the damage, so avoiding the damage is privately worthwhile.”

Do not add compensation to crop damage when calculating social cost. The damaged crop is a resource loss; compensation is a transfer between parties.

### Step 3: rancher has the right to let cattle roam

The farmer otherwise bears 1,000 of damage. If permitted to install the fence directly, the farmer is willing to spend 600 to prevent it. Alternatively, the farmer can pay the rancher to install it.

Write “Illustrative payment: 800.” Have students calculate gains relative to no fence:

Farmer’s gain =  $1,000 - 800 = 200$

Rancher’s gain =  $800 - 600 = 200$ .

Payments strictly between 600 and 1,000 make both parties better off in this bargain relative to no agreement. Endpoints leave one party indifferent. The exact division depends on bargaining; 800 is not a predicted unique payment.

**Ask:** “What stayed the same? What changed?” **Answer:** the fence is built in both rights cases; who pays and the division of gains change.

#### Step 4: change one number

Replace the fence cost with 1,400. Give students 30 seconds to choose the efficient outcome.

**Answer:** accept 1,000 of damage, assuming there is no cheaper remedy. Under farmer protection, the rancher compensates the farmer. Under the rancher's right to roam, the farmer bears the loss. Neither party can create a gain by spending 1,400 to prevent 1,000 of harm.

**Say:** "Bargaining can lead to an efficient outcome. That outcome may include damage."

Instructor cue: slide 19's statement that someone builds a fence needs the condition that the fence is worth its cost.

### 27–33 minutes | When bargaining is difficult — slide 20

State the theorem in simple language: "If rights are clear and enforceable, and bargaining is costless, parties can reach an efficient agreement." For this example, assume the parties can make the relevant payments and the assigned rights do not change the underlying costs or benefits.

Define **transaction costs**: finding the parties, negotiating, monitoring, and enforcing the agreement.

Return to the 600 fence and 1,000 damage. Add 500 in unavoidable negotiation and enforcement costs to the negotiated fence arrangement:

$$\text{Gain before transaction costs} = 400, \quad \text{gain after these costs} = 400 - 500 = -100.$$

**Say:** "This negotiated arrangement now costs more than it saves." This calculation assumes those extra costs are necessary for that arrangement; it does not rule out a cheaper direct solution.

**Ask:** "Why is a bargain harder with ten thousand affected residents?" Expected answers: identifying everyone, disagreement over harm and payment, holding out, relying on others to pay, and enforcement.

**Bridge:** "Negotiation depends on who can control access. We can use access and use to classify goods."

### 33–43 minutes | Board example 2: four types of goods — slides 21–27

Define the two terms one at a time:

**Excludable:** "It is feasible to prevent someone from using the good." Payment gates, tickets, and passwords are examples.

**Rival:** "One person's use leaves less quantity or quality for other people." Eating a particular sandwich is rival. Watching a broadcast does not use up the signal.

Draw an empty matrix. Ask both questions before writing a category name.

|               | Rival                                | Nonrival                                    |
|---------------|--------------------------------------|---------------------------------------------|
| Excludable    | Private good: sandwich               | Club good: uncrowded subscription broadcast |
| Nonexcludable | Common resource: open-access fishery | Public good: national defense               |

**Work aloud through the fishery:** “Can users be kept out under open access? No. Does my catch leave fewer fish for others? Yes. It is a common resource.”

**Work aloud through the broadcast:** “Can a provider require a subscription? Yes. Does another viewer consume the signal? No. It is a club good.”

Use slide 24. The intended answer is the free outdoor light display, assuming viewers cannot practically be excluded and additional viewers do not crowd each other out.

**Ask:** “Is a government-provided sandwich a public good?” **Answer:** no. The sandwich is still rival and excludable. Who provides it does not determine its category.

**Ask:** “Can a private good have a positive externality?” **Answer:** yes. An individual vaccination dose is rival and excludable, while reduced transmission can benefit others. Classification and spillovers are different questions.

## 43–50 minutes | Board example 3: one road, four cases — slides 26–28

Give pairs two minutes to fill the final column, then discuss. State that an enforced toll makes exclusion feasible and that an unrestricted road in this exercise has no effective access restriction.

| Road conditions         | Reasoning                                                    | Classification               |
|-------------------------|--------------------------------------------------------------|------------------------------|
| No toll, little traffic | No effective exclusion; another car creates negligible delay | Approximately public good    |
| Toll, little traffic    | Exclusion feasible; another car creates negligible delay     | Approximately club good      |
| No toll, congested      | No effective exclusion; another car delays others            | Common resource              |
| Toll, congested         | Exclusion feasible; another car delays others                | Private-good characteristics |

**Say:** “The physical road is the same. Access rules and congestion change its economic characteristics.”

Use slide 28. Expected textbook answer: club good. Add: “That describes access to an uncrowded gym. A machine with a queue is rival.” Accept a qualified answer supported by both tests.

Preview free riding: “If people cannot be excluded, they may wait for others to pay.” Save the numerical funding example for Class 3.

## 50–55 minutes | Exit check

Allow three minutes to write and two minutes to review.

1. A neighbor’s equipment causes 500 in annual harm. Soundproofing costs 200 annually and eliminates the harm. With costless bargaining, what is efficient? If the neighbor has the right to make noise, give one payment that benefits both parties relative to no agreement.
2. Classify an open-access fishery using both criteria.
3. Does assigning different rights necessarily change the efficient physical outcome in our bargaining model?

**Answers:** soundproofing; a payment of 350 from the affected person to the neighbor gives each a gain of 150. Fishery: nonexcludable and rival, hence common resource. Under the stated assumptions, changing rights changes distribution but not the efficient physical outcome.

**Closing sentence:** “Next time we will study why shared benefits can be underfunded and shared resources can be overused.”

## Pacing and optional depth

**If behind:** reduce the bicycle discussion to one minute and demonstrate two road cases instead of four. Keep the expensive-fence variation: it prevents a central misconception about bargaining.

**Optional extension:** add 400 in unavoidable transaction costs to the exit-check soundproofing bargain. The gross gain is 300, so that arrangement no longer creates a net gain. Ask whether a different institution could achieve the outcome more cheaply; students need not assume one always exists.

**Misconceptions to listen for:** efficient means fair; property rights guarantee a fence; compensation removes the physical harm; government provision makes something a public good; nonrival means unlimited capacity in all circumstances.